

A Pillar 6 electroweak-bridge programme note for TECT: first $\Delta F < 0$ broken-energy data point at $N = 16$ within the continuum-limit-scan protocol

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We present a candidate electroweak-bridge programme exploring the embedding of the BCC shell structure of the topological condensate into the Standard-Model three-generation + Higgs-doublet sector within Topological Energy Condensate Theory (TECT). Using the continuum-limit-scan protocol (Math235), we evaluate the candidate electroweak-bridge projector $\mathcal{P}_{EW}$ from the 12-dimensional BCC shell moduli space V_{shell} into the candidate 2-dimensional Higgs-doublet support V_{EW} (Math234). At lattice resolutions $N \in \{4, 8, 16, 32\}$, the broken-phase free-energy gap $\Delta F = F_{\text{broken}} - F_{\text{symmetric}}$ first turns negative at $N = 16$ with $\Delta F_{N=16} \approx -324.94$ in TECT internal units; the $N = 32$ value $\Delta F_{N=32} \approx -1021.7$ extends the trend but is *not* a continuum endpoint. The current canonical Pillar 6 tier is **T4 STRONG EVIDENCE**. Under the Math292 four-gauge acceptance criterion (G1 finite- f ; G2 broken-energy $\Delta F < 0$; G3 transverse-Hessian PSD; G4 extraction OK), the present data point fulfils G2 only at $N = 16$ and remains pending on G1/G3/G4 per the Math310-AddA re-certification programme. Math293 false-negative taxonomy and the Math294/Math294-AddA basin-of-attraction analyses motivate $N = 16$ as the empirical first-entry resolution but do not, at this tier, certify embedding. A pre-registered F-Pillar6 falsification gate (deadline **2026-05-29**) governs the next T4 $\rightarrow$ T6 promotion attempt; the present document does *not* certify Pillar 6 closure.

I. INTRODUCTION

The Standard Model's three-fermion-generation + Higgs-doublet structure is, within most frameworks, an external phenomenological input. TECT poses a different question: can the structure be made *compatible* with a topological-condensate vacuum that supports the requisite gauge embedding? The present paper does not assert that the answer is positive at theorem level. It reports the first numerical data point at which the candidate broken phase becomes energetically favoured ($\Delta F < 0$) within the continuum-limit-scan protocol, and audits the data point against the Math292 four-gauge acceptance criterion.

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The BCC condensate is not uniform; its 12 nearest-neighbour generators form a crystallographic shell. Each shell mode is a candidate degree of freedom for the embedding programme; the projection of the 12-dimensional shell moduli onto a candidate 2-dimensional Higgs-doublet support is the object of study (Math234). The present paper does *not* prove that the three SM generations correspond to specific shell modes, nor that the Yukawa hierarchy is fixed; both remain Pillar 6 **T1 OPEN** (Math234 sub-tasks SO(10)-rep and Yukawa-extraction).

The contribution of this paper is purely the ΔF data point and its honest audit against Math292 acceptance: the broken phase exists at $N \geq 16$, the embedding remains pending.

II. SETTING AND MAIN RESULT

The BCC shell is described by the moduli vector $\phi = (\phi_1, \dots, \phi_{12}) \in V_{\text{shell}}$, where each $\phi_i \in \mathbb{C}$ is the amplitude on one BCC nearest-neighbour direction. The order parameter is

$$\Psi(\mathbf{r}) = \sum_{i=1}^{12} \phi_i \exp(i\mathbf{q}_i \cdot \mathbf{r}), \quad (1)$$

with $\mathbf{q}_i$ the 12 BCC nearest-neighbour wave vectors.

The candidate electroweak-bridge projector $\mathcal{P}_{\text{EW}} : V_{\text{shell}} \rightarrow V_{\text{EW}} \cong \mathbb{C}^2$ is defined by coefficient decomposition (Math234):

$$H = \mathcal{P}_{\text{EW}} \cdot \phi = \begin{pmatrix} \alpha_H \\ \beta_H \end{pmatrix}, \quad \alpha_H = \sum_{j \in S_u} c_j^\alpha \phi_j, \quad \beta_H = \sum_{j \in S_d} c_j^\beta \phi_j, \quad (2)$$

where S_u, S_d partition the 12 directions, and $c_j^{\alpha/\beta}$ are structure constants fixed by the BCC lattice geometry. The map (2) is a candidate projector; whether it identifies the physical Higgs doublet of the SM is the embedding question that remains open.

Theorem 1 (First $\Delta F < 0$ data point at $N = 16$ (**T4 STRONG EVIDENCE**, G2-of-4-gauges only)). *Under hypotheses*

(H1) *the continuum-limit-scan protocol of Math235–236;*

(H2) *the Math292 four-gauge acceptance criterion interpreted in the present paper at the G2 (broken-energy entry) gauge only, i.e. $\Delta F < 0$, with G1 (finite f), G3 (transverse-Hessian PSD), G4 (extraction OK) marked pending per Math310-AddA;*

(H3) *the Math293 false-negative taxonomy classifying $N < 16$ data as sub-critical;*

(H4) the Math294/Math294-AddA basin-of-attraction analyses identifying $N = 16$ as the empirical marginal-basin resolution,

the broken-phase free-energy gap evaluated at $N = 16$ is

$$\boxed{\Delta F_{N=16} \approx -324.94 \pm 8.7 \quad (\text{TECT units; } G2 \text{ satisfied; } G1/G3/G4 \text{ pending}).} \quad (3)$$

Remark 2 (Honest scope). (3) is a single-resolution, single-gauge data point within the Math292 four-gauge framework. *It does not certify embedding.* Promotion to T6 PROVED CONDITIONAL requires all four Math292 gauges to pass simultaneously at $N \geq 16$, plus discharge of the Math310-AddA transverse-projection re-certification. The F-Pillar6 falsification gate (deadline 2026-05-29) governs the next T4 $\rightarrow$ T6 promotion attempt.

III. METHOD: CONTINUUM-LIMIT SCAN AND THE MATH292 FOUR-GAUGE AUDIT

a. Step 1 (continuum-limit-scan protocol). Math235 defines a four-stage protocol implemented in Math236 [?]: (i) discretise the BCC shell on lattices with $N \in \{4, 8, 16, 32\}$; (ii) minimise the free-energy functional $F[\phi]$ numerically on each grid; (iii) compute $\Delta F = F_{\text{broken}} - F_{\text{symmetric}}$ as a function of N ; (iv) extrapolate to the continuum ($N \rightarrow \infty$) via a quadratic fit, conditional on at least three super-critical data points and a verified residual.

b. Step 2 (Math292 four-gauge acceptance criterion). The Math292 acceptance gate for an embedding-data point is the *conjunction* of four conditions:

$$\text{Math292-Accept} \iff \underbrace{(f > 0)}_{\mathbf{G1}} \wedge \underbrace{(\Delta F < 0)}_{\mathbf{G2}} \wedge \underbrace{(\lambda_{\min}^{\text{transverse}} \geq 0)}_{\mathbf{G3}} \wedge \underbrace{(\text{extraction} = \text{OK})}_{\mathbf{G4}}. \quad (4)$$

The label *A-valid* that appeared in earlier drafts of this paper referred only to G2; that usage was an over-claim and is corrected here. The present data point passes G2 at $N = 16$ only and is pending on G1, G3, G4 per the Math310-AddA re-certification programme.

c. Step 3 (false-negative taxonomy). Math293 classifies sub-critical data ($\Delta F \geq 0$ at $N < 16$ due to finite-size artefact) as false negatives that must not block the search. Conversely, $\Delta F < 0$ at $N \geq 16$ counts as a candidate positive provided the remaining Math292 gauges are independently verified [?].

d. Step 4 (basin-of-attraction analyses). Math294 and Math294-AddA analyse the marginal basin of the broken phase and identify $N = 16$ as the empirical first-entry resolution. This justifies $N = 16$ as the operating point of the present audit; it does not promote the data point past T4 in the absence of the G1/G3/G4 verifications.

e. Step 5 (numerical audit). Math290 documents wrapper and solver bugs encountered in the first production run, with corrections applied to produce the values reported here [?].

IV. NUMERICAL RESULTS AND MATH292 GAUGE AUDIT

The continuum-limit-scan results for ΔF in TECT internal units are summarised in Table I.

N	ΔF (numerical)	Phase	Math292 verdict
4	+127.3	symmetric	G2 fails (sub-critical)
8	+45.2	symmetric	G2 fails (sub-critical)
16	-324.94	broken	G2 only; G1/G3/G4 pending
32	-1021.7	broken	G2 only; G1/G3/G4 pending

TABLE I. Continuum-limit-scan data for the broken-phase free-energy gap. Only G2 (broken-energy entry) of the Math292 four-gauge acceptance criterion is satisfied at $N = 16, 32$; the remaining gauges G1, G3, G4 await Math310-AddA re-certification.

Remark 3 ($N = 32$ is not a continuum endpoint). The $N = 32$ value $\Delta F \approx -1021.7$ extends the broken-phase trend but is *not* a continuum-limit verdict. A continuum-limit extrapolation requires (i) a third super-critical data point at $N \geq 64$, (ii) a quadratic-fit residual below the per-point numerical-error budget, and (iii) the Math292 gauges G1/G3/G4 verified at every super-critical N in the fit range. None of (i)–(iii) is discharged by the present data set; the $N = 32$ value is therefore reported as a plateau-trending data point and *not* as a confirmed endpoint.

The candidate Higgs-doublet support (α_H, β_H) extracted at $N = 16$ inherits its symbolic structure from Math234. Whether this extraction reproduces the physical SM Higgs charge and hypercharge assignments is a separate Math234-extraction sub-task and is, at the present canonical tier, **T1 OPEN**.

V. DISCUSSION AND NEXT STEPS

The present paper records a single empirical fact: within the continuum-limit-scan protocol of Math235–236, the broken-phase free-energy gap ΔF first turns negative at lattice resolution $N = 16$, with values $\Delta F_{N=16} \approx -324.94$ and $\Delta F_{N=32} \approx -1021.7$ in TECT internal units. The data point passes the G2 (broken-energy entry) gauge of the Math292 four-gauge acceptance criterion

only, and remains pending on G1/G3/G4. It does not, at the present canonical tier, certify the candidate electroweak-bridge embedding programme.

Residual tasks before T4 $\rightarrow$ T6 promotion:

1. **Math310-AddA transverse-projection re-certification** discharging G3 ($\lambda_{\min}^{\text{transverse}} \geq 0$);
2. independent verification of G1 (finite f) and G4 (extraction OK) at $N \in \{16, 32\}$;
3. a third super-critical data point at $N \geq 64$ with Math292 four-gauge passes, enabling the continuum-limit quadratic fit;
4. separate Math234-extraction sub-task verifying the candidate Higgs-doublet identification.

The pre-registered F-Pillar6 falsification gate carries deadline **2026-05-29**. Success on all four tasks supports the T4 $\rightarrow$ T6 promotion attempt; partial success keeps the result at T4 with explicit caveat enumeration.

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- J. Lee, TECT-Math236: continuum-limit-scan numerical implementation, TECT internal note (2026).
 - J. Lee, TECT-Math293: false-negative taxonomy for sub-critical N, TECT internal note (2026).
 - J. Lee, TECT-Math290: continuum-scan wrapper / solver bug audit, TECT internal note (2026).